

Rahul Bangar

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SUMMARY

Software Development Engineer with 1.5+ years of experience designing, building, and operating large-scale distributed systems and microservices on AWS. At Amazon OpenSearch, delivers customer-facing features end-to-end in Java and Go, improves reliability and performance of production services, and builds monitoring, alarming, and infrastructure-as-code (AWS CDK/CloudFormation). Strengths in distributed systems, cloud infrastructure, REST APIs, observability, CI/CD, and data structures & algorithms.

EDUCATION

National Institute of Technology, Tiruchirappalli

Bachelor of Technology

Computer Science & Engineering

Jul 2020 – Jul 2024

EXPERIENCE

Amazon (AWS OpenSearch)

Software Development Engineer

Feb 2025 – Present

- Delivered two customer-facing index settings (custom timestamp field for time-series collections and hot-to-warm storage tiering) end-to-end across 9 Java/Go microservices, feature-flagged with a staged beta-to-prod rollout and validated across 38 test scenarios; exposed via REST APIs.
- Eliminated STS AssumeRole throttling on the bulk-delete path by refactoring AWS service clients into cached singletons, cutting throttling errors from ~3,400 per run to zero and reducing collection-drain time by up to 30%.
- Consolidated IAM inline policies from 8 to 2 (~70% smaller) to clear the 10 KB limit and unblock a failing production CloudFormation (AWS CDK) deployment; fixed HTTP 500 errors on bulk deletes by batching DynamoDB reads to the 100-item limit.
- Authored observability and monitoring policies in Go for the cluster-manager fleet, covering JVM memory pressure, node-role mismatch, and cluster fault-detection, each shipped with CloudWatch alarms and a staged global rollout.
- Remediated a fleet-wide upgrade blocker across production OpenSearch domains (largest: 117 nodes / 10,611 shards) by building a cross-region detection tool and authoring an operational runbook.

Accenture

Advanced App Engineering Analyst

September 2024 – January 2025

- Completed intensive Big Data training (Hadoop, Hive, Cassandra, MySQL) and applied it to hands-on data-pipeline exercises.

Accenture

Advanced App Engineering Analyst Intern

June 2023 – July 2023

- Led vulnerability assessments in SAP systems, detecting and documenting critical security gaps.
- Proposed strategic mitigation solutions that decreased average vulnerability resolution time by 25%.
- Communicated findings with stakeholders, improving the team's understanding of secure development processes.

HACKATHONS

TransfiNITTe – Intra NITT Hackathon

Nov 2023

- Secured 2nd place (out of ~100 teams) building a Llama-2 chatbot to bridge startup engineering and customer support communication gaps.
- Implemented GitHub webhook-based backend for real-time repository data analysis.

PROJECTS

MPQUAD (Multipath Quad Protocol)

Nov 2022

- Designed **MPQuad**, an innovative protocol tailored for **multi-path** communication within **TCP**.
- The experimental analysis showcased increased throughput and reduced queuing delays.
- Presented project findings at **3rd International Conference on Advances in Computing, Communication, Embedded and Secure Systems (ACCESS) in 2023**, highlighting the protocol's potential for improving network efficiency and reliability.
- Published research paper on MPQuad in **IEEE**, contributing to academic discourse in the field.

Mini C Compiler

Apr 2023

- Built front-end and back-end modules supporting complex control flow in C language, including nested structures.
- Implemented six major compilation phases, deepening expertise in parsing and code generation.

Utilizing Blockchain for decentralized IoT Authentication

Jan 2024 - Apr 2024

- Deployed custom chaincode to establish authentication protocols within the **Hyperledger Fabric network**.
- Implemented the chaincode logic in **Golang**, comprising **CRUDE** functionality of devices.
- Created the **backend server** using **Gin**, a lightweight web framework in Golang.
 - It facilitates seamless communication between the user interface and the Hyperledger Fabric for efficient handling of HTTP requests and responses, enhancing the overall performance and scalability of the system.
- Applied a robust authentication mechanism allowing authorized devices to interact with the network securely. Remarkably reduced authentication delay from **14 seconds to 2-3 seconds**.
- Integrated **Hyperledger Explorer** as an interface between the blockchain network and the administrative users.
 - This allowed administrators to visualize and interact with the blockchain data in a user-friendly manner, enhancing transparency and accessibility.
- Integrated the chaincode and backend server components, conducting thorough testing to ensure the reliability, security, and scalability of the system.

SKILLS

- **Programming:** Java, Python, Go, C, C++, JavaScript, TypeScript, Dart, HTML, CSS
- **Frameworks/Tools:** Gin, Flutter, CI/CD, Bash, Git
- **Databases:** DynamoDB, MySQL, Cassandra, Hive
- **Cloud:** AWS (Lambda, EC2, S3, DynamoDB, CloudWatch, Step Functions, IAM/STS, CloudFormation/CDK), Hyperledger Fabric
- **Other:** Data Structures, Algorithms, System Design, Distributed Systems, Microservices, REST APIs, Unit/Integration Testing, DevOps, LaTeX

Updated on Aug 07, 2026